

Certified Prime-Gated Universal Multiplicity Equation (CP-UME)

This document formalizes the Universal Multiplicity Equation (UME) into the **Certified Prime-Gated Universal Multiplicity Equation (CP-UME)**, suitable for rigorous analysis, simulation, and falsifiable evaluation.

It contains:

1. **Implementation Checklist** (author-facing, one-page style)
 2. **Drop-in Manuscript Section** (spec-level text with equations)
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1) One-Page Implementation Checklist

1.1 Interpretation Lock: Two Lanes (Choose One Per Model Instance)

Template (dimensional):

$$\frac{\partial \psi}{\partial t} = A(t)\psi + B(t) \int_0^t K(t-\tau) I(\tau) d\tau + \Gamma(t) \mathcal{T}(\psi, \psi) + D(t) \mathcal{D}\psi + H(t) \mathcal{N}(\psi) + \xi(t, \cdot).$$

- ψ : state (vector field)
- $\mathcal{T}$: bounded bilinear operator (tensor coupling)
- $\mathcal{N}$: nonlinearity (e.g., componentwise power)
- (K) : memory kernel (set $K \equiv 1$ only if you truly want a raw integral; see Gripenberg, Londen & Staffans, 1990)
- $\mathcal{D}$: smoothing/diffusion operator (lane-dependent)
- ξ : forcing/noise

Lane A — Physical PDE.

- Domain: $x \in \Omega \subset \mathbb{R}^d$
- State: $\psi(t, x) \in \mathbb{R}^K$ (or $\mathbb{C}^K$)
- Operator: $\mathcal{D} = \Delta_x$
- **Must declare physical meaning of ψ** (density/amplitude/order parameter) to fix units.

Lane B — Graph/ML diffusion.

- Domain: graph $G = (V, E)$, $N = |V|$
- State: $\psi(t) \in \mathbb{R}^{K \times N}$ (or flattened $\mathbb{R}^{KN}$)
- Operator: $\mathcal{D} = -L_G$, where L_G is a specified graph Laplacian (normalized or unnormalized; state which) (e.g., Chung, 1997; Belkin & Niyogi, 2003; Belkin, Niyogi & Sindhwani, 2006).
- **Must declare adjacency/kernel construction** (concept graph, neural connectivity, similarity graph).

Rule: Present the template once, then instantiate Lane A and Lane B separately. Do not mix lanes within proofs.

1.2 Nondimensionalization (Mandatory; Fixes Dimensional Issues; Enables Numerics)

Choose characteristic scales:

- amplitude ψ_0
- time t_0
- (Lane A) length x_0

Define dimensionless variables:

$$\phi = \psi / \psi_0, \quad \tilde{t} = t / t_0, \quad \tilde{x} = x / x_0 \quad \text{(Lane A)}.$$

Rewrite the equation **entirely** in dimensionless form and use that form for **all** analysis/simulation.

Typical scaling identities:

- $\tilde{A}(\tilde{t}) = t_0 A(t_0 \tilde{t})$
 - Lane A diffusion: $\tilde{D}(\tilde{t}) = \frac{t_0}{x_0^2} D(t_0 \tilde{t})$
 - Lane B diffusion (if L_G dimensionless): $\tilde{D}(\tilde{t}) = t_0 D(t_0 \tilde{t})$
 - Bilinear coefficient: $\tilde{\Gamma}(\tilde{t}) = t_0 \psi_0 \Gamma(t_0 \tilde{t})$
 - Power nonlinearity $\mathcal{N}(\psi) = \psi^{\odot n}$: $\tilde{H}(\tilde{t}) = t_0 \psi_0^{n-1} H(t_0 \tilde{t})$
 - Noise: $\tilde{\xi}(\tilde{t}, \cdot) = \frac{t_0}{\psi_0} \xi(t_0 \tilde{t}, \cdot)$
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1.3 Constrained Prime Gating (Parsimonious + Identifiable)

Let $P_N = \{2, 3, 5, \dots, p_N\}$ be the first N primes.

For each coefficient operator/matrix $C(t) \in \{A, B, \Gamma, D, H\}$:

$$C(t) = \sum_{p \in P_N} c_p(t) W_p.$$

Requirements:

- $\{W_p\}$ are **declared** and **linearly independent** (preferably orthonormal under Frobenius/Hilbert-Schmidt inner product).

Constraints (must appear in the model definition):

1. **Finite support:** only primes in P_N .
2. **Decay envelope:** $|c_p(t)| \leq C p^{-\alpha}$ (use $\alpha > 1$ if taking $N \rightarrow \infty$ in theory).

3. **Uniform ℓ^1 bound:** $\sum_{p \in P_N} |c_p(t)| \leq C_*$ for all t . (Analogous to ℓ^1 sparsity control; see Tibshirani, 1996.)
 4. (Optional) **Temporal smoothness:** $|\dot{c}_p(t)| \leq \nu p^{-\alpha}$.
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1.4 Falsifiability Bridge (Prime vs Matched Baselines)

Claim to test: prime gating yields stability/structure not achieved by equally sparse non-prime gating.

Required gating schemes (matched parameter count and matched decay envelope):

- **P (Prime):** P_N
- **R (Random sparse):** R_N integers, same size, same decay envelope
- **H (Harmonic/log-spaced):** e.g., $\{1, 2, 4, 8, \dots\}$ or log-spaced integers
- **S (Scrambled primes):** primes kept but permute mapping $p \mapsto W_p$

Required metrics:

- **Stability margin/basin size:** fraction of initial conditions with bounded trajectories over horizon $[0, T]$
- **Largest Lyapunov exponent:** $\lambda_{\max}$ (Benettin et al., 1980; Wolf et al., 1985).
- **Scaling with N :** trends of $\lambda_{\max}(N)$, basin size vs N

Decision rule:

- If P does not beat R/H/S reproducibly on at least one primary metric, reframe as “structured sparse spectral gating,” not prime-specific structure.
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1.5 Fast Validation (2–5 Days)

A (1–2 days): Minimal simulation

- Lane A: 1D PDE on $[0, 1]$ with Neumann BCs, or Lane B: small graph $N \approx 50$
- Use $K = 3$
- Sweep $(\alpha, C_*, \tilde{D})$
- Report basin size + $\lambda_{\max}$ for P/R/H/S

B (2–3 days): ML task

- Implement discretized CP-UME as a recurrent cell
- Train on chaotic prediction (Lorenz) or long-horizon memory task (Bengio, Simard & Frasconi, 1994).
- Report gradient explosion incidence + test error for P/R/H/S

C: Only if A and B show prime-specific advantage, proceed to quantum/tunneling framing tests.

2) Drop-in Manuscript Section (Spec-Level)

3. Certified Prime-Gated Universal Multiplicity Equation (CP-UME)

3.1 Template Model and Two-Lane Instantiation

We define a general dynamical template for a K -component state ψ with linear drift, memory, bilinear coupling, smoothing/diffusion, nonlinearity, and forcing:

$$\partial_t \psi = A(t)\psi + B(t) \int_0^t K(t-\tau) I(\tau) d\tau + \Gamma(t) \mathcal{T}(\psi, \psi) + D(t) \mathcal{D}\psi + H(t) \mathcal{N}(\psi) + \xi(t, \cdot). \quad \text{tag{1}}$$

Here $\mathcal{T}$ is a bilinear operator and $\mathcal{N}$ is a nonlinear map (e.g., componentwise power). The smoothing operator $\mathcal{D}$ is chosen according to one of two instantiations:

Lane A (Physical PDE). Let $\Omega \subset \mathbb{R}^d$ and $\psi(t, \cdot) \in L^2(\Omega; \mathbb{R}^K)$ (or $\mathbb{C}^K$). We set $\mathcal{D} = \Delta_x$ and interpret ψ as a physical field (e.g., density, amplitude, order parameter), which fixes units.

Lane B (Graph/ML). Let $G = (V, E)$ be a graph with $N = |V|$ and $\psi(t) \in \mathbb{R}^{K \times N}$. We set $\mathcal{D} = -L_G$, where L_G is a specified graph Laplacian constructed from an explicitly defined adjacency/kernel. (e.g., Chung, 1997; Belkin & Niyogi, 2003; Belkin, Niyogi & Sindhwani, 2006).

All analyses and experiments are performed within one lane at a time; Lane A and Lane B are separate instantiations of the shared template (1).

3.2 Nondimensionalization (Primary Form)

To ensure dimensional consistency and numerical stability, we nondimensionalize (1). Choose characteristic scales: amplitude ψ_0 , time t_0 , and (Lane A) length x_0 . Define

$$\phi = \psi / \psi_0, \quad \tilde{t} = t / t_0, \quad \tilde{x} = x / x_0 \quad \text{for Lane A.} \quad \text{tag{2}}$$

In Lane A, $\Delta_x = x_0^{-2} \Delta_{\tilde{x}}$. Writing all terms in $(\tilde{t}, \tilde{x})$ yields the dimensionless evolution

$$\partial_{\tilde{t}} \phi = \tilde{A}(\tilde{t}) \phi + \tilde{B}(\tilde{t}) \int_0^{\tilde{t}} \tilde{K}(\tilde{t} - \tilde{\tau}) \tilde{I}(\tilde{\tau}) d\tilde{\tau} + \tilde{\Gamma}(\tilde{t}) \mathcal{T}(\phi, \phi) + \tilde{D}(\tilde{t}) \mathcal{D}\phi + \tilde{H}(\tilde{t}) \mathcal{N}(\phi) + \tilde{\xi}(\tilde{t}, \cdot). \quad \text{tag{3}}$$

Typical coefficient scalings are

$$\tilde{A}(\tilde{t}) = t_0 A(t_0 \tilde{t}), \quad \tilde{\Gamma}(\tilde{t}) = t_0 \psi_0 \Gamma(t_0 \tilde{t}), \quad \text{tag{4}}$$

$$\tilde{D}(\tilde{t}) = \begin{cases} \frac{t_0}{x_0^2} D(t_0 \tilde{t}), & \text{Lane A,} \\ D(t_0 \tilde{t}), & \text{Lane B if } L_G \text{ is dimensionless,} \end{cases} \quad \tilde{\xi}(\tilde{t}, \cdot) = \frac{t_0}{\psi_0} \xi(t_0 \tilde{t}, \cdot). \quad \text{tag{5}}$$

For a componentwise power nonlinearity $\mathcal{N}(\psi) = \psi^{\odot n}$, we take $\tilde{\mathcal{N}}(\phi) = \phi^{\odot n}$ and define

$$\tilde{H}(\tilde{t}) = t_0 \psi_0^{\wedge\{n-1\}} H(t_0 \tilde{t}), \quad \text{tag{6}}$$

absorbing all dimensional dependence into $\tilde{H}$. Equation (3) is therefore dimensionally consistent by construction and is used throughout.

3.3 Certified Prime-Mixture Coefficients (Prime Gating)

Let $P_N = \{2, 3, 5, \dots, p_N\}$ denote the first N primes. For each coefficient operator/matrix $C(\tilde{t}) \in \{\tilde{A}, \tilde{B}, \tilde{\Gamma}, \tilde{D}, \tilde{H}\}$, we impose the constrained prime-mixture representation

$$C(\tilde{t}) = \sum_{p \in P_N} c_p(\tilde{t}) W_p, \quad \text{tag{7}}$$

where $\{W_p\}_{p \in P_N}$ are fixed, declared, and linearly independent basis operators (preferably orthonormal under the Hilbert–Schmidt/Frobenius inner product). The scalar weights satisfy

- **Finite support:** only $p \in P_N$.
- **Decay envelope:** $|c_p(\tilde{t})| \leq C p^{-\alpha}$ (use $\alpha > 1$ for asymptotic theory).
- **Uniform ℓ^1 bound:** $\sum_{p \in P_N} |c_p(\tilde{t})| \leq C_*$ for all $\tilde{t}$.
- *(Optional)* **Temporal smoothness:** $|\dot{c}_p(\tilde{t})| \leq \nu p^{-\alpha}$.

These constraints ensure prime gating is a parsimonious, identifiable modulation rather than a generic function approximator.

3.4 Well-Posedness (Theorem-Ready Conditions)

Let $\mathcal{H}$ be the relevant Hilbert space (Lane A: $L^2(\Omega; \mathbb{R}^K)$; Lane B: $\mathbb{R}^{K \times N}$). Define the linear operator

$$\tilde{\mathcal{L}}(\tilde{t}) := \tilde{A}(\tilde{t}) + \tilde{D}(\tilde{t}), \quad \tilde{\mathcal{H}} := \mathcal{H}.$$

Assume:

1. $\tilde{\mathcal{L}}(\tilde{t})$ generates an evolution family $U(\tilde{t}, \tilde{s})$ with $\|U(\tilde{t}, \tilde{s})\| \leq M e^{-\omega(\tilde{t}-\tilde{s})}$.
2. $\mathcal{T}$ is bounded bilinear: $\|\mathcal{T}(u, v)\| \leq b_* \|u\| \|v\|$.
3. $\tilde{\mathcal{N}}(\tilde{t}, \cdot)$ is locally Lipschitz on bounded sets.
4. Prime-mixture constraints (7) hold.

Theorem 1 (Local well-posedness). Under assumptions 1–4, for any $\phi_0 \in \mathcal{H}$ there exists $T > 0$ and a unique mild solution $\phi \in C([0, T]; \mathcal{H})$ to (3). If, additionally, linear dissipation dominates nonlinear growth (standard energy/small-gain conditions), the solution is global. See, e.g., Pazy (1983), Henry (1981), Temam (1988), and Evans (1998).

3.5 Falsifiable Predictions and Mandatory Baselines

CP-UME makes a prime-specific empirical claim:

Claim (Prime-specific stability). Under matched parameter budgets and matched decay envelopes, prime gating yields increased stability (larger basin of attraction and/or more negative $\lambda_{\max}$) compared to non-prime sparse gating.

Mandatory baseline gating schemes (matched N , matched decay envelope):

- **P:** primes P_N
- **R:** random sparse integers R_N
- **H:** harmonic/log-spaced integers (e.g., $\{1, 2, 4, 8, \dots\}$)
- **S:** scrambled primes (permute mapping $p \mapsto W_p$)

Primary metrics:

1. Stability margin (bounded trajectory fraction over $[0, T]$).
2. Largest Lyapunov exponent $\lambda_{\max}$.
3. Scaling with N : trends of $\lambda_{\max}(N)$ and basin size vs N .

A positive result requires P to exceed R/H/S reproducibly on at least one primary metric; otherwise the contribution should be reframed as structured sparse spectral gating.

3) Notation Defaults (Recommended)

- Use ψ for dimensional state, ϕ for dimensionless.
- Use $\tilde{t}, \tilde{x}$ for dimensionless variables.
- Use $\mathcal{D} = \Delta_x$ (Lane A) and $\mathcal{D} = -L_G$ (Lane B).
- Use $\odot$ for componentwise powers.

4) Minimal Experimental Protocol (Appendix-ready)

- Fix $K = 3$, choose a simple bounded $\mathcal{T}$ (e.g., $[\mathcal{T}(u, v)]_k = \sum_{j,l} T_{kjl} u_j v_l$ with bounded T_{kjl}).
- Choose $N \in \{5, 10, 20\}$ and compare P/R/H/S at each N .
- Report (i) basin size estimate, (ii) $\lambda_{\max}$, (iii) scaling curves vs N .

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